JAVA ENTERPRISE PRACTICALS

Practical 6: Expression Language in JSP

Environment: Apache NetBeans · GlassFish Server · JDK 21

1. Aim

To demonstrate the use of Expression Language (EL) in JSP to access request attributes and form parameters without writing scriptlet code for output.

2. Theory

Expression Language (EL) is a simplified language used in JSP to access data stored in JavaBeans, request attributes, session attributes, and request parameters. EL expressions are written using the syntax `${expression}`.

In this practical:

- Form data is submitted from `index.html` to `e1Demo.jsp`.
- The JSP stores the values as request attributes using scriptlets.
- EL is used to display the attribute values (`${studentName}`, etc.).
- The `param` implicit object is used to access form parameters directly (`${param.name}`, etc.).

3. Complete Steps

1. Open Apache NetBeans IDE.
2. Select **File** → **New Project**.
3. Select **Java with Ant** → **Web** → **Web Application** (or the equivalent Web Application template available in your NetBeans version). Click **Next**.
4. Enter the project name as **JSPExpressionLanguage** and select the project location.
5. Select GlassFish Server and a compatible Java EE/Jakarta EE version. Click **Finish**.
6. In the Projects panel, expand the project and open the **Web Pages** folder. In a Maven project, use `src/main/webapp`.
7. Create an HTML file named `index.html` and enter the first program given below.
8. Create a JSP file named `e1Demo.jsp` in the same web root folder and enter the second program.
9. Save both files.
10. Right-click the project and select **Run**. Wait for GlassFish Server to deploy the application.
11. Open `index.html` in the browser using the project URL, for example: `http://localhost:8080/JSPExpressionLanguage/index.html`
12. Enter the Name, Age and Course details in the form.
13. Click **Submit**. The browser will open `e1Demo.jsp` and display the student details using EL attributes and the `param` implicit object.

4. Complete Code

4.1 index.html

Create this file under Web Pages (or `src/main/webapp`).

```
<!DOCTYPE html>

<html>
    <head>
        <title>Expression Language</title>
        <meta charset="UTF-8">
```

```
        <meta name="viewport"
              content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <form action="e1Demo.jsp" method="post">
            Name:
            <input type="text" name="name">
            <br><br>

            Age:
            <input type="number" name="age">
            <br><br>

            Course:
            <input type="text" name="course">
            <br><br>

            <input type="submit" value="Submit">
        </form>
    </body>
</html>
```

4.2 e1Demo.jsp

Create this JSP file in the same web root folder.

```
<%@page language="java"
        contentType="text/html"
        pageEncoding="UTF-8" %>

<%
    request.setAttribute("studentName",
            request.getParameter("name"));

    request.setAttribute("studentAge",
            request.getParameter("age"));

    request.setAttribute("studentCourse",
            request.getParameter("course"));
%>

<!DOCTYPE html>

<html>
    <head>
        <title>JSP Page</title>
        <meta charset="UTF-8">
        <meta name="viewport"
              content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <h2>Student Details</h2>

        Name: ${studentName}
        <br><br>

        Age: ${studentAge}
        <br><br>

        Course: ${studentCourse}
        <br><br>

        <hr>

        <h3>Using param Implicit Object</h3>

        Name: ${param.name}
        <br><br>

        Age: ${param.age}
```

```
        <br><br>

        Course: ${param.course}
    </body>
</html>
```

5. Important Code Explanation

Code / Concept	Purpose
action="e1Demo.jsp"	Sends the form data to the JSP page.
method="post"	Submits form data using HTTP POST.
request.setAttribute(...)	Stores a value as a request-scoped attribute.
${studentName}	EL expression that reads the request attribute studentName.
${param.name}	EL expression that reads the form parameter named name.
param implicit object	Gives direct access to request parameters in EL.
<%@page ... %>	Page directive that configures the JSP page.

6. Expected Output

After entering sample values (e.g. Name: **Rahul**, Age: **21**, Course: **BCA**) and clicking Submit, the JSP page displays:

```
Student Details

Name: Rahul

Age: 21

Course: BCA

---------------------------------------

Using param Implicit Object

Name: Rahul

Age: 21

Course: BCA
```

Both sections show the same values: the first uses request attributes set in the scriptlet, and the second uses the `param` implicit object.

7. Key Points to Remember

1. EL expressions use the syntax `${...}`.
2. `${studentName}` reads a request attribute named studentName.
3. `${param.name}` reads the form parameter named name.
4. The `param` object is one of the EL implicit objects.
5. EL avoids writing Java code for simple output in the JSP page.
6. Request attributes are set using `request.setAttribute()` in a scriptlet.

8. Result

A simple web form was created that submits Name, Age and Course to a JSP page. The JSP uses Expression Language to display the values both as request attributes and via the param implicit object, demonstrating basic EL usage in JSP.